Vishnuhemanth Tiruvalluru

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.20
Narayana Junior College	Class XII (Board of Intermediate Education, Andhra Pradesh)	2017	97.50%
Sai Sree Margadarsi English Medium High School	Class X (Board Of Secondary Education Andhra Pradesh)	2015	9.30%

THESIS & RESEARCH ACTIVITES

Forecasting Financial Connectedness Networks

(Under Journal review)

- Developed deep learning models to forecast **Diebold-Ylmaz connectedness matrices**, representing the evolving **transmission of financial shocks across markets**.
- Tackled the problem of forecasting **structured, interpretable financial networks** instead of individual asset prices, preserving meaningful spillover relationships.
- Implemented and benchmarked deep learning models against classical forecasting methods on real-world financial datasets.

AI for Finance, Economies & Society (AIFES) Lab:

- Part of the developing team in trustworthy AI-driven solutions in finance by addressing research and industry-oriented problems.
- Delivered **30+ technical presentations** on Math finance, Stochastic Processes, Stochastic Calculus and AI for Finance.
- Collaborate with faculty and researchers through Literature review, Technical discussions, and ongoing Research initiatives.

Synthetic Financial Data creation(Ongoing):

- Create synthetic data in a controllable environment to implement stylised facts of real world financial data
- Control the connectedness between two assets in market and verify forecasting connectedness.

PROJECTS

Language Modeling for Romanized Telugu WhatsApp Text

- Built a next-word prediction model for **Code-Mixed Telugu** using **296K+ WhatsApp messages**.
- Fine-tuned GPT-2 with **LoRA** and evaluated the impact of custom tokenization and multilingual pretraining.

Improving Federated Learning under Data Heterogeneity

- Proposed an adaptive loss weighting framework to reduce distribution shift caused by heterogeneous client data.
- Combined loss decomposition with quality-aware model aggregation, improving classification accuracy by **1-2%** compared with the baseline.

RELEVANT COURSES

Matrix Theory, Probability and Random Variables, Foundations of Machine Learning, Deep Learning, Convex Optimization, Stochastic Process, AI in Finance

SKILLS

Mathematical Finance: Probability, Stochastic Processes, Stochastic Calculus, Derivative Pricing & Hedging, Portfolio Optimization, Implied Volatility

Mathematics: Probability, Stochastic Processes, Stochastic Calculus

Programming: Python, C, C++

Frameworks: Scikit-learn, PyTorch

ML and DL: Regression, RNN, LSTM

EXPERIENCE

Associate System Engineer - Trainee

Tata Consultancy Services (TCS) | July 2021 - July 2022

Worked as an Associate System Engineer Trainee, gaining practical exposure to software development processes, collaborative project environments, and automation workflows.

POSITION OF RESPONSIBILIY

Student Organizer - India AI Pre Summit IIT Hyderabad

- One of the student main organizer for India AI Pre summit happened in IIT Hyderabad Conclave on Inclusion for Social Empowement on Jan 16 2026

Volunteer | Prayas Student Volunteering Club

- Contributed to outreach initiatives promoting the importance of education among underprivileged students and their families, encouraging awareness and access to educational opportunities.

PG Buddy | Sunshine

- Mentored and supported first-year postgraduate students, helping them navigate academic and campus life to ensure a smooth transition into the institute.

HOBBIES & INTERESTS

Cricket, Listening songs, Cooking, Food Exploration